Chapter 5

Data Analysis

Ashim Khadka

Research Techniques

1. Data Analysis

2. Types of Data Analysis
   - Quantitative Data Analysis
     - Types of quantitative data
   - Data Coding
   - Data Presentation
   - Analysis Techniques
     - Descriptive Statistics
     - Inferential Statistics
     - Regression Analysis
     - Time Series Analysis
     - Multivariate Analysis

3. Qualitative Data Analysis
   - Qualitative Data
   - Analyzing Textual Data
   - Analyzing Non-Textual Data

Data Analysis

- process of systematically applying statistical and logical techniques to describe, organize, summarize, and interpret collected data
- helps researchers identify patterns, test hypotheses, and derive meaningful conclusions to support decision-making and theory-building
- In research, data analysis allows scholars to:
  - Extract meaningful insights from collected data.
  - Identify patterns and trends to support or refute research questions.
  - Ensure reliability and accuracy in their findings.
  - Make informed decisions based on statistical evidence.

Purpose of Data Analysis in Research

- Summarization helps convert raw data into meaningful insights by organizing it into structured formats, such as: Tables, Graphs, Charts, and Statistical summaries (mean, median, mode, standard deviation, etc.)
  - Example: A researcher collects student test scores from 10 schools to analyze learning outcomes. Instead of listing thousands of scores, they can summarize:
    - Mean score: The average performance of students.
    - Median score: The middle value to avoid extreme values affecting results.
    - Standard deviation: The variation in student performance
- researchers must **interpret** the results to derive meaningful conclusions
  - Identifying patterns, correlations, or anomalies.
  - Comparing results against the hypothesis.
  - Understanding the real-world significance of data.

- Validation ensures that the research findings are accurate, reliable, and free from errors or biases
  - Checking for missing data (e.g., incomplete survey responses)
  - Verifying data consistency (e.g., ensuring all ages are within a reasonable range)
  - Cross-checking data sources (e.g., validating online data with official reports)
- Hypothesis testing is used to determine if research findings are statistically significant
  - Formulating a null hypothesis (H_0) (assumption that there is no effect)
  - Testing against an alternative hypothesis (H_1) (assumption that there is an effect)
  - Using statistical tests (e.g., t-test, chi-square test, ANOVA)
- Data-driven decision-making is crucial in research
  - Draw conclusions that answer research questions
  - Provide recommendations for future research or policy changes
  - Improve existing processes based on evidence

Steps in Data Analysis

1. Data Collection - Gather data through experiments, surveys, or observations
2. Data Cleaning - Remove missing values, inconsistencies, or outliers
3. Data Processing - Convert data into a structured format
4. Exploratory Data Analysis (EDA) - Identify trends, distributions, and patterns
5. Statistical Analysis - Use statistical tests to validate hypotheses
6. Data Interpretation - Draw conclusions based on results
7. Reporting - Present findings through tables, graphs, and research papers

Types of Data Analysis

1. Quantitative Data Analysis
2. Qualitative Data Analysis

Quantitative Data Analysis

- involves numerical data and statistical methods to uncover patterns, relationships, and trends
- includes understanding data types, coding for analysis, presenting data visually, and applying statistical techniques
- data generated by experiments and surveys
- descriptive statistical techniques allow more patterns to be found such as the average value of a set
- Types of quantitative data:
  1. Discrete Data
  2. Continuous Data
  3. Nominal Data
  4. Ordinal Data
  5. Interval Data
  6. Ratio Data

Discrete Data

- Consists of distinct, separate values (e.g., number of students in a class: 20, 21, 22)
- Cannot take intermediate values
- Example: Number of cars in a parking lot
- Analysis: Suitable for frequency tables and bar charts

Continuous Data

- Can take any value within a range (e.g., height: 1.75m, 1.76m, etc.)
- Measured rather than counted
- Example: Temperature readings
- Analysis: Use histograms, mean, and standard deviation

Nominal Data

- Categorical data without order (e.g., gender: male, female)
- Used for labeling variables
- Example: Eye color categories
- Note: Cannot calculate mean; use mode or frequencies

Ordinal Data

- Categorical data with a natural order (e.g., satisfaction level: low, medium, high)
- Differences between categories are not necessarily equal
- Example: Rating scales (1-5 stars)
- Note: Can calculate median, but mean may be inappropriate; suitable for non-parametric tests like Mann-Whitney U test

Interval Data

- Numerical data with equal intervals but no true zero (e.g., temperature in Celsius)
- Example: IQ scores, calendar dates
- Note: Can calculate mean and standard deviation, but ratios are meaningless (e.g., 20°C is not twice as hot as 10°C)

Ratio Data

- Numerical data with equal intervals and a true zero (e.g., weight, time)
- Example: Height in meters, income in dollars
- Note: Allows for all statistical calculations, including ratios (e.g., 10 kg is twice as heavy as 5 kg)

Data Coding

- process of converting raw data into a format suitable for statistical analysis, often by assigning numerical values to categories
- Example: If a survey has a question on satisfaction level
  - Very Satisfied $\rightarrow$ 5
  - Satisfied $\rightarrow$ 4
  - Neutral $\rightarrow$ 3
  - Dissatisfied $\rightarrow$ 2
  - Very Dissatisfied $\rightarrow$ 1
- conversion allows statistical analysis using numerical methods
- Avoid duplicating codes or mislabeling categories
- Handling missing data: delete or data imputation

Steps involved in Data Coding

1. Identify Variables: Determine what data needs coding (e.g., responses to a survey with categorical options)
2. Assign Codes:
   - For nominal data: Male = 1, Female = 2.
   - For ordinal data: Low = 1, Medium = 2, High = 3.
3. Create a Codebook: Document the coding scheme for reference, ensuring consistency (e.g., "1 = Male, 2 = Female").

Data Presentation

- Once data is collected, it needs to be visualized effectively
- Visuals make data easier to interpret and communicate, aiding in identifying patterns or trends
- Types:
  - Tables
  - Bar charts
  - Scatter Graph
  - Line graph

Tables

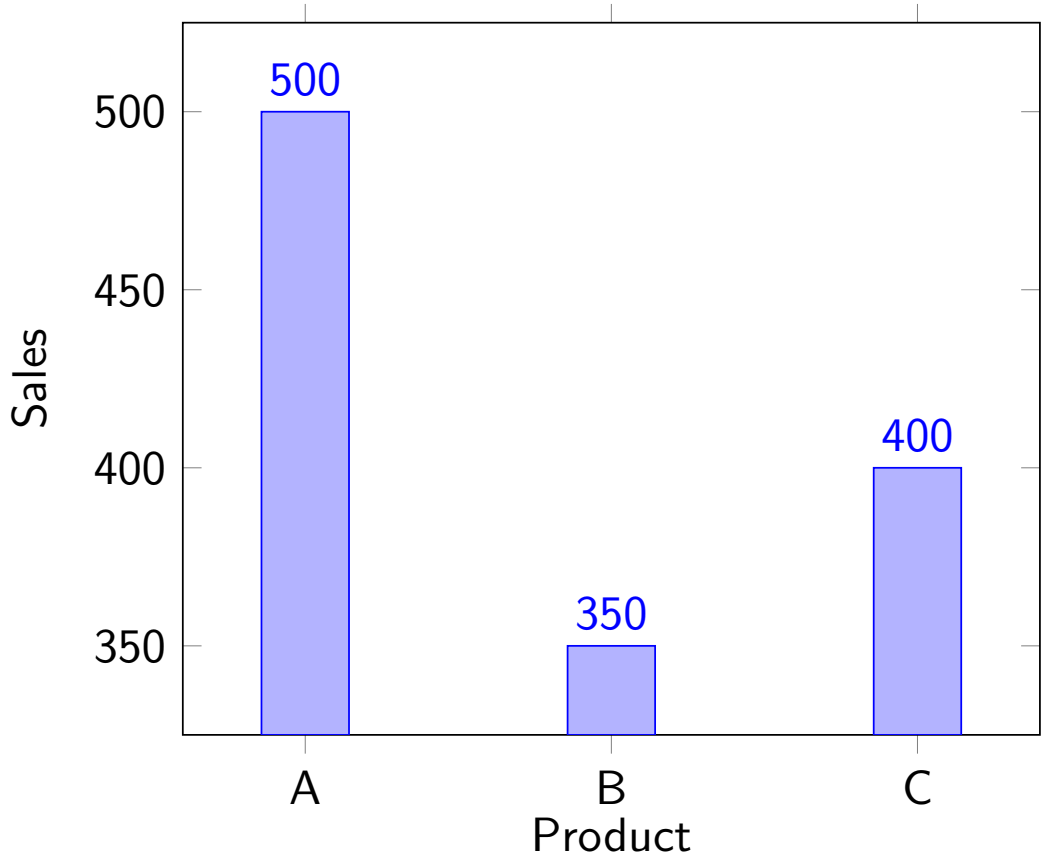
- Organize data in rows and columns for exact values
- Best for showing detailed data
- Use: Compare multiple variables or present raw data

Table: Sales of Different Products

Product	Sales
A	500
B	350
C	400

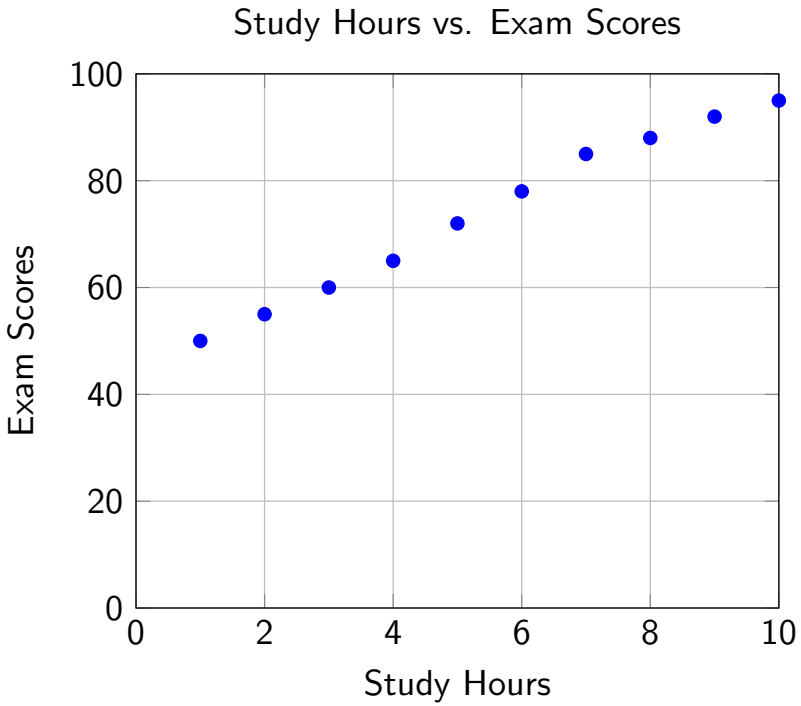
Bar Charts

- Use bars to represent categorical data
- Vertical or horizontal; gaps between bars indicate discrete categories
- Note: Preferred over pie charts for comparing categories due to easier visual comparison



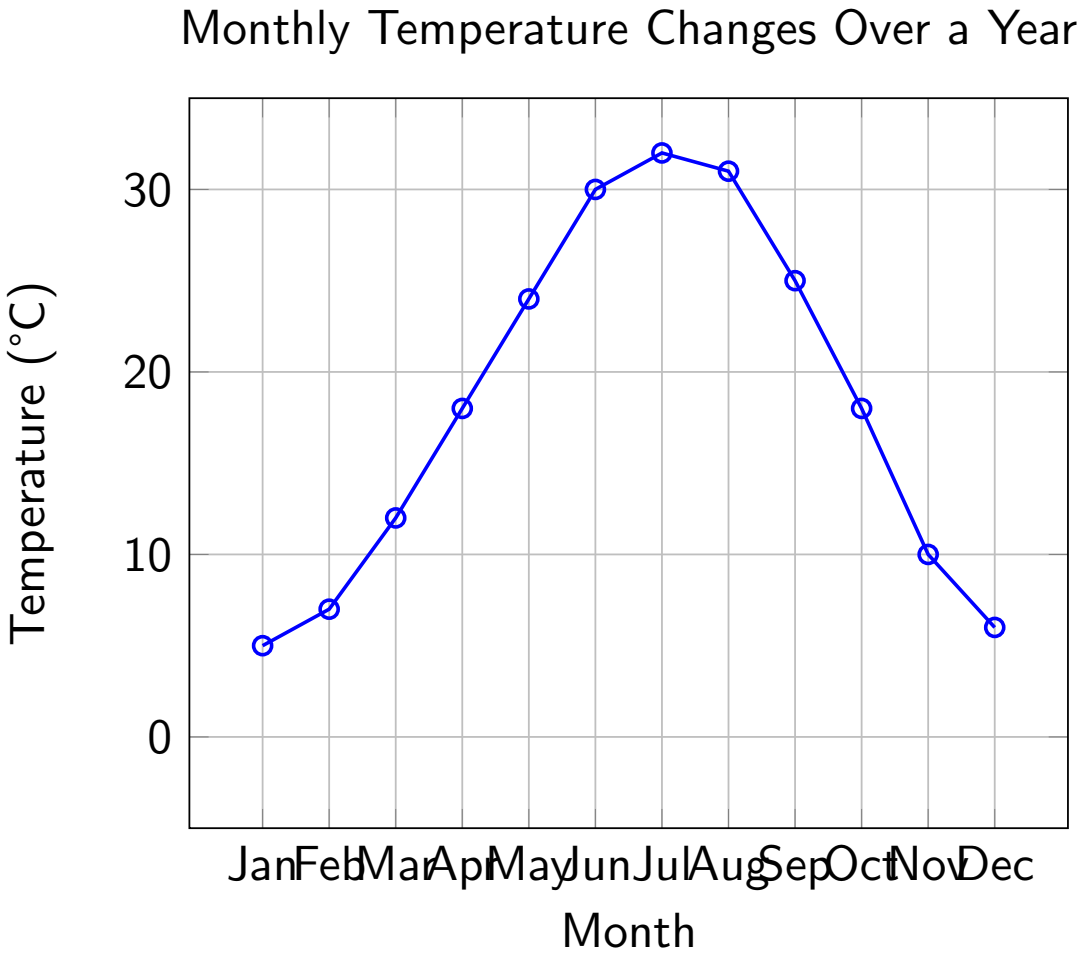
Scatter Graph

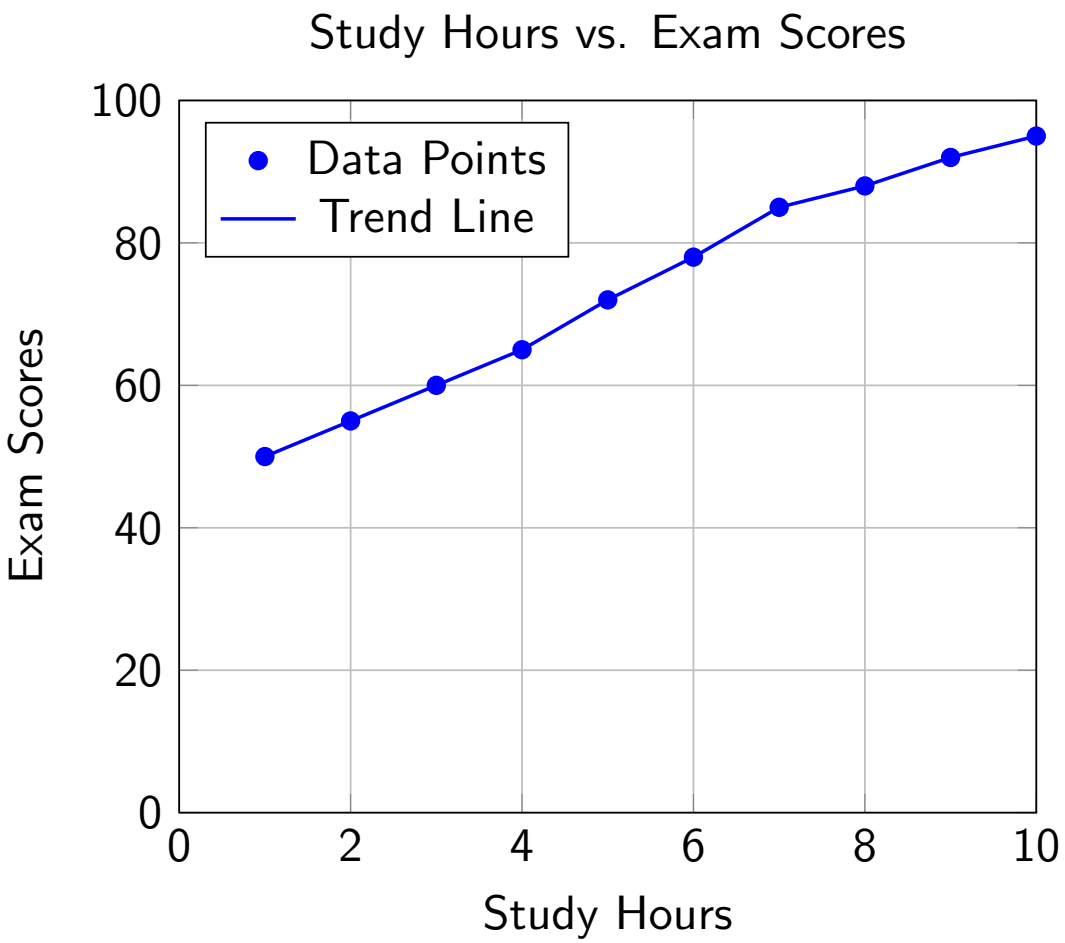
- Used to show relationships between two numerical variables
- Plot individual data points on X-Y axes to show relationships (e.g., height vs. weight)
- Useful for correlation analysis, identifying trends or outliers
- Example: Plot study hours vs. exam scores to see if more study correlates with higher scores



Line Graph

- Best for continuous data, especially time series
- Connect data points to show trends
- Example: Show monthly temperature changes over a year




Study Hours vs. Exam Scores
Data Points
Trend Line
Exam Scores
Study Hours
0
20
40
60
80
100
0
2
4
6
8
10

Analysis Techniques

1. Descriptive Statistics
2. Inferential Statistics
3. Regression Analysis
4. Time Series Analysis
5. Multivariate Analysis

Descriptive Statistics

- Summarize data to describe central tendency and variability
  - Use: Understand the dataset's basic characteristics without inference.
- Measures: Mean (average), median (middle value), mode (most frequent), standard deviation (spread), variance, range
  - Mean is sensitive to outliers; median is more robust for skewed data
- Example: Average test score $= 75$, standard deviation $= 10$, indicating scores vary by 10 points around the mean.

Inferential Statistics

- Draw conclusions about a population from a sample, testing hypotheses
- Used to make predictions or test hypotheses
  - Z-test: Compares means when the population variance is known
  - T-test: Compares means of two groups
  - Chi-square test: Analyzes categorical data
  - ANOVA: Compares means of multiple groups

Z-test

- Large samples, known population parameters/variance
- Compare sample mean to the population mean when the population variance is known
- Example: Test if a new drug improves recovery time, assuming known population recovery time variance

$$Z = \frac{(\bar{X} - \mu)}{\sigma/\sqrt{n}}$$

where $\bar{X}$: sample mean, μ : population mean, σ : population standard deviation, n : sample size

T-test

- Compare means of two groups (e.g., male vs. female test scores) when the population variance is unknown or the sample size is small ($n < 30$)
- relies on t-distribution
- helps to determine if differences between groups are statistically significant

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where $\bar{X}_1, \bar{X}_2$: sample means, s_1^2, s_2^2 Sample variances, n_1, n_2 : sample sizes

Types of T-Tests

1. One-Sample T-Test:
   - Compares the mean of a sample to a known population mean
   - Example: Testing if the average IQ of students in a class ($\bar{X}$ differs from the national average (μ)
2. Independent Two-Sample T-Test (Unpaired):
   - Compares the means of two independent groups
   - Example: Comparing male vs. female exam scores
3. Paired T-Test (Dependent Samples):
   - Compares means before and after an intervention on the same group
   - Example: Measuring student scores before and after using a new teaching method

ANOVA (Analysis of Variance)

- For Multiple Groups
  - More than two groups, to avoid multiple t-tests
- Compare means across multiple groups
- Examples: test scores by teaching method: lecture, group work, online

Chi-square test

- For Categorical Data
- Used to test relationships between two categorical variables
  - gender and voting preference
- Types: Test of independence (are variables related?), goodness of fit (does data fit expected distribution?)

Regression Analysis

- Model relationships between dependent and independent variables.
- Types:
  - Linear regression ($y = mx + b$, e.g., predict sales from advertising spend)
  - multiple regression ($y = b_0 + b_1x_1 + b_2x_2$, e.g., predict grades from study hours and attendance)
- Example: Predict student grades based on study hours, showing a positive slope (more study, higher grades)
- Use: Prediction, understanding variable relationships, assuming linearity

Time Series Analysis

- Analyze data collected over time (e.g., monthly sales trends)
- Components: Trend (long-term direction), seasonality (regular patterns, e.g., holiday sales spikes), noise (random variation)
- Example: Forecast next quarter's revenue using past sales, identifying seasonal peaks
- Use: Forecasting, trend analysis, requires time-ordered data

Multivariate Analysis

- Analyze multiple variables simultaneously to find patterns or relationships
- Techniques: Factor analysis (reduce variables to underlying factors), cluster analysis (group similar observations), discriminant analysis (classify into groups)
- Example: Group customers by income, age, and spending habits using cluster analysis for targeted marketing
- Use: Complex datasets, high dimensionality, to uncover hidden structures

Qualitative Data Analysis

- Qualitative data analysis focuses on non-numerical data, such as text or images, to uncover themes, meanings, and patterns
- nature of qualitative data and techniques for analyzing textual and non-textual forms

Qualitative Data

- Qualitative data is descriptive, non-numerical data that captures qualities, opinions, or behaviors (e.g., interview transcripts, photos, videos)
- Subjective and context-dependent, reflecting human experiences
- Rich in detail but harder to quantify compared to numerical data
- Sources: Interviews, focus groups, observations, open-ended survey responses, diaries, visual materials

Comparison with Quantitative

- Quantitative answers "how much?" or "how many?" (e.g., 70% passed the exam)
- Qualitative answers "why?" or "how?" (e.g., "Students felt the exam was fair")
- Complementary: Often used together for mixed-methods research

Analyzing Textual Data

- Learn systematic approaches to analyze text-based qualitative data
- Collect data by gathering transcripts, notes, or documents, ensuring ethical considerations (e.g., consent, anonymity)
- Read and re-read data to understand content, immersing in the context
- Assign labels to segments of text to identify themes or patterns (e.g., "stress" for "I feel overwhelmed by deadlines")
- Group codes into broader themes (e.g., "emotional responses" including stress, anxiety, satisfaction)
- Explain findings in context, linking to research questions or theoretical frameworks

Techniques: Analyzing Textual Data

- Content Analysis: Quantify frequency of themes (e.g., count how often "cost" is mentioned in customer reviews)
- Thematic Analysis: Identify recurring patterns or themes (e.g., fear of failure in student interviews)
- Narrative Analysis: Focus on stories or sequences, analyzing the structure and meaning (e.g., a patient's journey through treatment)

Analyzing Non-Textual Data

- Explore methods to analyze qualitative data beyond text, such as images or videos
- Types:
  - Images: Photos, drawings, advertisements (e.g., marketing visuals, protest photos)
  - Audio/Video: Recordings of interviews, focus groups, or events, capturing tone, gestures, expressions
  - Objects: Artifacts or physical items (e.g., historical tools, cultural artifacts)

Analysis Process

1. Description: Detail what is observed (e.g., in a photo, note bright colors, smiling faces, urban setting)
2. Contextualization: Link to broader context (e.g., cultural significance, historical period, social norms)
3. Interpretation: Infer meaning (e.g., a smile in a photo suggests happiness, but consider cultural variations)

Techniques

1. Visual Analysis: Examine composition, symbols, colors (e.g., red may symbolize danger or passion, depending on context).
2. Semiotics: Study signs and meanings, analyzing how visuals communicate (e.g., a logo's design conveys brand identity).
3. Discourse Analysis: Analyze spoken or visual communication patterns, considering power dynamics or narratives